Setup Steps

Architecture

The project runs in four parts on one machine, each in its own terminal window.

Part	What it is	Runs with	Address
`infra`	PostgreSQL database + Keycloak login server	Docker	Keycloak: http://localhost:8180 · Postgres: port `5432`
`backend`	Spring Boot REST API	Maven + Java 21	http://localhost:8080/swagger-ui.html
`storefront`	Customer shopping site (React)	Node.js	http://localhost:5173
`admin`	Admin portal (React)	Node.js	http://localhost:5174

```
apparel-commerce/
├── infra/        # docker-compose.yml for Postgres + Keycloak
├── backend/      # Spring Boot API (Flyway migrations, Swagger UI)
├── storefront/   # React customer site
└── admin/        # React admin portal
```

Tested versions

The project has been verified on this setup (Windows 11, build 10.0.26300.9974):

Tool	Version	Check with
Java (Eclipse Temurin JDK)	21.0.13 LTS (build 21.0.13+9)	`java -version`
Apache Maven	3.9.16	`mvn -version`
Node.js	v24.16.0	`node -v`
Docker	29.9.3 (build 5457199)	`docker --version`
WSL	3.0.5.0 (kernel 6.18.58.1-1)	`wsl --version`
Git	any recent version	`git --version`
Docker Compose	v2 (bundled with Docker Desktop)	`docker compose version`

Java **21** is required. Maven must run on Java 21, so check that the `Java version:` line in `mvn -version` shows `21.x`.

Prerequisites (one-time install, Windows)

Do these in order, and restart the PC where noted.

1. **Java 21.** Install Eclipse Temurin JDK 21 (x64 `.msi`) from adoptium.net. On the options screen, enable **Set JAVA_HOME variable** and **Add to PATH**. It installs to `C:\Program Files\Eclipse Adoptium\jdk-21.0.13.9-hotspot`, not to `C:\Program Files\Java`.
2. **Maven.** Download `apache-maven-3.9.x-bin.zip` from maven.apache.org and extract it, for example to `C:\Users\<you>\Downloads\maven\apache-maven-3.9.16`.
3. **Node.js.** Install the LTS version from nodejs.org with default options.
4. **Git.** Install from git-scm.com with default options.
5. **Windows features for Docker.** In PowerShell as Administrator, run each command separately:

   ```
   dism.exe /online /enable-feature /featurename:VirtualMachinePlatform /all
   /norestart
   dism.exe /online /enable-feature /featurename:Microsoft-Windows-Subsystem-Linux
   /all /norestart
   ```

   Each should end with "The operation completed successfully." Restart the PC afterwards.
6. **WSL.** From github.com/microsoft/WSL/releases, download the `.x64.msi` under **Assets** of the latest release and install it. Restart the PC.
7. **Docker Desktop.** Install it from docker.com, open it once, and wait until it shows **Engine running**. You can skip signing in.
8. **JAVA_HOME.** In admin PowerShell:

   ```
   [Environment]::SetEnvironmentVariable("JAVA_HOME", "C:\Program Files\Eclipse
   Adoptium\jdk-21.0.13.9-hotspot", "Machine")
   ```
9. **Path.** Open Start → *Edit the system environment variables* → Environment Variables → System variables → `Path` → Edit. Add these if they're missing, with no leading spaces:
   - `%JAVA_HOME%\bin` (move it to the top)
   - `C:\Users\<you>\Downloads\maven\apache-maven-3.9.16\bin`

   Then close and reopen your terminal.
10. **Verify.** Open a new terminal and run the commands in Tested versions.

Project setup (one time)

```
cd C:\Users\<you>\apparel-commerce

cd storefront
npm install

cd ..\admin
npm install
```

npm may report "vulnerabilities". That's normal for local development. Don't run `npm audit fix --force`, because it can break the project.

Optional: AI chatbot key

The chatbot needs a real Anthropic API key from console.anthropic.com, which is billed per use. Without a key, everything except the chatbot works. **Keep the key secret and never commit it to Git.**

Running the project (every day)

Start Docker Desktop and wait for **Engine running**. Then open four terminals and leave each one running.

#	Folder	Command	Ready when
1	`infra`	`docker compose up -d`	`docker compose ps` shows both services running and http://localhost:8180 loads (30–60 s)
2	`backend`	`mvn spring-boot:run`	You see `Started CommerceApplication`
3	`storefront`	`npm run dev`	http://localhost:5173 opens
4	`admin`	`npm run dev`	http://localhost:5174 opens

Demo admin login: `admin@loomandco.example` / `admin123` (for local development only)

Enabling the chatbot

Set the key in terminal 2 **before** starting the backend:

```
:: Command Prompt
set ANTHROPIC_API_KEY=your-real-key
```

```
# PowerShell
$env:ANTHROPIC_API_KEY = "your-real-key"
```

Stopping

1. Press `Ctrl+C` in terminals 2, 3 and 4.
2. In terminal 1 (`infra`), run `docker compose stop`. Your data is kept.

Resetting the database to demo data

```
cd infra
docker compose down -v
docker compose up -d
```